

Amith, here's a walkthrough of every code section in your KRAS project and what each one accomplishes within the broader analysis.

1. Reproducing the Heatmaps (Figure 2A & 2B) — LoVo and HCT-116 Cell Lines

This section replicates Figure 2 from a published Nature paper comparing gene expression between KRAS-mutant and wild-type colorectal cancer cell lines.

Finding the 600 DEGs (Differentially Expressed Genes) — LoVo version:

```
lovo_df = pd.read_csv("LoVo vs control.csv")
deg600_df = pd.read_csv("600 DEGS.csv")
matched_df = lovo_df[lovo_df['Ensemble Gene ID'].isin(deg600_df['EnsembleGeneID'])]
matched_df.to_csv("Matched_LoVo_600_DEGs.csv", index=False)
```

This loads the full LoVo differential expression results and a curated list of 600 DEGs identified in the original study. It then filters the LoVo dataset to keep only genes that appear in that 600-gene list using `.isin()` — essentially an inner join on Ensembl Gene IDs. The output is a CSV of matched genes with their expression values across replicates. The purpose is to isolate the biologically meaningful genes so the heatmap isn't cluttered with tens of thousands of rows.

Heatmap with Dendrogram — LoVo version:

```
expression_cols = ['SW48-2', 'SW48-1', 'SW48-3', 'LoVo-2', 'LoVo-1', 'LoVo-3']
scaler = StandardScaler()
z_scores = scaler.fit_transform(expr_df.T).T
```

This selects the expression columns for the wild-type (SW48) and mutant (LoVo) replicates, then applies Z-score normalization row-wise (per gene). The `.T` transpose trick is needed because `StandardScaler` normalizes columns by default — transposing, scaling, and transposing back makes it operate on rows instead. Z-scoring puts every gene on the same scale (mean=0, std=1), so highly expressed genes don't visually dominate the heatmap.

The dendrogram is computed via `linkage(z_df, method="complete", metric="euclidean")`, which performs hierarchical clustering on genes using complete-linkage (maximum pairwise distance between clusters). This dendrogram is plotted on the left side of the heatmap to show which genes have similar expression patterns. The

heatmap itself uses a blue-white-red (**bwr**) colormap centered at zero, so blue = downregulated, red = upregulated relative to that gene's mean.

The HCT-116 version is structurally identical, just swapping LoVo for HCT-116 and adding a 4th HCT-116 replicate column. The key biological takeaway your team draws from both panels is that despite being different cell lines, the shared G13D KRAS mutation produces similar expression profiles — visible as consistent red/blue clustering patterns across both heatmaps.

2. Downloading and Processing GTEx Healthy Tissue Data (Step 1)

```
filename = "gene_tpm_v10_pancreas.gct.gz"
df = pd.read_csv(filename, sep='\t', skiprows=2)
tpm_mean = tpm_values.mean(axis=1)
result = pd.concat([gene_names, tpm_mean.rename("mean_TPM")], axis=1)
```

GTEx provides gene expression data in GCT format (a tab-delimited matrix with two header lines). **skiprows=2** skips those headers. The file contains TPM values for every gene across dozens of individual pancreas samples. This code averages all those samples into a single **mean_TPM** column per gene — collapsing individual variation into a representative "healthy pancreas" expression profile. This gets repeated for colon and lung tissues. These averages become the baseline you compare cancer cell lines against.

3. Extracting Cancer Cell Line Data from DepMap (Step 2)

```
target_ids = {"ACH-000982"}
for chunk in pd.read_csv("OmicsExpressionProteinCodingGenesTPMLogp1.csv", index_col=0,
    chunksize=1000):
    filtered = chunk.loc[chunk.index.intersection(target_ids)]
```

The DepMap expression matrix is enormous (thousands of cell lines × ~19,000 genes), so this uses chunked reading to avoid memory issues. It reads 1,000 rows at a time, checks if any match the target DepMap ID (e.g., ACH-000982 for the GP2D colon cancer line), and collects matches. The three cell lines chosen — GP2D (colon), PANC0403 (pancreas), T3M10 (lung) — all carry the KRAS G12D mutation, which is the specific variable being tested. The expression values are already log2(TPM+1) transformed in DepMap.

4. Filtering Top 1000 Genes (Step 3)

```
top_genes = df.loc[cell_line].sort_values(ascending=False).head(1000)
```

For each cancer cell line, this sorts all ~19,000 genes by expression level and keeps the top 1,000 most highly expressed. This is a dimensionality reduction step — rather than comparing all genes (most of which are noise or housekeeping), you focus on the genes that are most actively transcribed in the cancer cells. These are the genes most likely to reveal biology driven by the KRAS mutation.

5. Merging Cancer vs. Healthy Tissue (Step 5)

```
cell_line_df["Gene"] = cell_line_df["Gene"].str.extract(r"^([^\(]+)").iloc[:, 0]  
comparison_df = pd.merge(cell_line_df, tissue_df[["Description", "mean_TPM"]],  
                          left_on="Gene", right_on="Description", how="inner")
```

This is where the analysis connects cancer to healthy tissue. The regex `r"^([^\(]+)"` strips parenthetical annotations from gene names (like Entrez IDs) so they match the GTEx naming convention. Then it performs an inner merge — only genes present in *both* the cancer cell line's top 1000 and the healthy tissue dataset survive. This gives you paired TPM values: one from the cancer cell, one from the healthy tissue, for each gene. You repeat this for all three tissue types.

6. PCA Clustering Analysis (Step 7)

```
merged = pd.concat([colon_df, pancreas_df, lung_df], axis=1, join="inner")  
expression_matrix = merged.T  
pca = PCA(n_components=2)  
pc = pca.fit_transform(expression_matrix)
```

This is the core analysis answering your research question. After merging all six samples (3 cancer lines + 3 healthy tissues) on their shared genes, transposing puts samples as rows and genes as features. PCA reduces ~hundreds of gene dimensions into 2 principal components that capture the most variance. The scatter plot then shows whether cancer cell lines cluster together (suggesting KRAS mutation drives a shared transcriptomic program) or cluster with their tissue of origin (suggesting tissue identity dominates). Your result — cancer lines clustering together — supports the conclusion that KRAS G12D imposes a common expression signature across tissue types.

7. Hierarchical Clustering and Heatmap (Step 8)

```
dendrogram(linkage(expression_matrix, method="ward"), labels=expression_matrix.index)
sns.clustermap(expression_matrix, cmap="vlag", standard_scale=1, col_cluster=False)
```

This provides a complementary view to the PCA. Ward's method minimizes within-cluster variance when merging clusters, producing a dendrogram that quantifies how "distant" each sample is from the others. The clustermap visualizes the full expression matrix as a heatmap with hierarchical clustering on the sample (row) axis but *not* on genes (`col_cluster=False`), so you can directly see which samples have similar expression patterns. `standard_scale=1` normalizes columns (genes) to [0,1] range for visualization. This confirms the PCA finding with a different method.

8. Log2 Fold Change Bar Plots (Step 9)

```
fc = expression_matrix.loc[cancer] - expression_matrix.loc[tissue]
top_fc = fc.abs().sort_values(ascending=False).head(20)
```

For each cancer/tissue pair, this computes the log2 fold change by subtracting the healthy tissue's log2(TPM+1) from the cancer cell line's value. Because the data is already log-transformed, subtraction in log-space equals division in linear space — so this gives you fold change directly. It then picks the 20 genes with the largest absolute fold change and plots them as bar charts. These are the genes most differentially expressed between the KRAS-mutant cancer and its tissue of origin, many of which (MIF, UBE2C, TOP2A, CDC20, RRM2, CCNB1) are known downstream targets or effectors of KRAS signaling, which validates the biological relevance of your analysis.

In summary, the code pipeline flows from data acquisition (GTEx, DepMap) → filtering and normalization → merging cancer vs. healthy → dimensionality reduction and clustering → fold change identification. Each step builds toward answering whether KRAS-mutant cancers share a common transcriptomic signature, and the answer your data supports is yes — the mutation, not the tissue of origin, is the dominant factor driving expression patterns.